

Aggregate Class Cheat Sheet

m `Aggregate(name, exp_el=0, exp_premium=0, exp_lr=0, exp_en=0, exp_attachment=None, exp_limit=inf, sev_name="", sev_a, sev_b, sev_mean, sev_cv, sev_loc, sev_scale, sev_xs, sev_ps, sev_wt, sev_lb, sev_ub, sev_conditional, occ_reins, occ_kind, freq_name, freq_a, freq_b, freq_zm, freq_p0, agg_reins, agg_kind, approximate, note, hints)`^[0]

The constructor follows the DecL clauses (exposure, severity, occurrence reinsurance, frequency, aggregate reinsurance). Usually built via `build('agg ...')`. The following tables show all

m methods, and fields or properties (used interchangeably). Comments elucidate the meaning of more obscure entries.

1. Specification & creation

`name`, `limit`, `attachment`, `frequency`, `sevs` (list of Severity), `en`, `n` (frequency), `value_type` (loss / premium for pnl), `sev_pick_attachments`, `sev_pick_losses`, `note`, `hints`, `program`, `pprogram` (pretty), `pprogram_html`, `spec`, `spec_ex`

2. Update

`log2`, `bs` (bucket size), `sev_calc` (round / forward / backward), `discretization_calc` (distribution / survival / both), `normalize`, `padding`, `approximate` (exact / sgamma / slognorm), `fzapprox`, `dsev_bucket`, `reins_bucket`, `x_min`, `x_max`, `i0`, **m** `update`, **m** `update_work`, **m** `discretize`, **m** `recommend_bucket`, **m** `picks`, **m** `rescale` (homog. sev / inhomog. freq), **m** `snap`, **m** `unwrap`

3. Moments

est prefix = estimated from FFT
`agg_m`, `agg_cv`, `agg_sd`, `agg_var`, `agg_skew`, `est_m`, `est_cv`, `est_sd`, `est_var`, `est_skew`, `sev_m`, `sev_cv`, `sev_sd`, `sev_var`, `sev_skew`, `est_sev_m`, `est_sev_cv`, `est_sev_sd`, `est_sev_var`, `est_sev_skew`, **m** `freq_pmf`

4. Statistical functions

m `cdf`, **m** `sf`, **m** `pdf`, **m** `pmf`, **m** `q` (lower quantile = VaR), **m** `ppf`, **m** `tvar`, **m** `sev` (exact severity cdf/sf/pdf), **m** `q_sev`, **m** `tvar_sev`, **m** `var_dict`^[1], **m** `sample`

5. Validation

`valid`, `validation_eps`, `describe` (validation stats), **m** `explain_validation`, **m** `aggregate_error_analysis` (agg error over bs), **m** `severity_error_analysis` (trunc. / discr. by component), `bs_window_df`, `bs_description`, `bs_explanation`, `tail_class`, `tail_df`, `tail_description`, `tail_explanation`

6. Output dataframes

`density_df`^[1] (main output), `stats_df`, `info`, `density`, `agg_density`, `sev_density`, `sev_density_df`, `ftagg_density` (FT), `xs`, `xs_sev`, `figure`, bounded *see also Reinsurance*

7. Reinsurance

`occ_kind`, `occ_reins`, `agg_kind`, `agg_reins`, `agg_density_gross/ceded/net`, `sev_density_gross/ceded/net`, `occ_ceder`, `occ_netter`, `agg_ceder`, `agg_netter`, **m** `apply_occ_reins`, **m** `apply_agg_reins`, **m** `reins_description`, `reins_kinds`, `reins_density_df`, `reins_describe`, `reins_stats_df`

8. Visualization

m `plot`, **m** `reins_occ_plot`, **m** `occ_bivariate`, `figure`

9. Risk and pricing

m `price`, **m** `price_pentagon`, **m** `apply_distortion`, **m** `cramer_lundberg` (aka pollaczec_khinchine; prob. of eventual ruin)

10. Approximations

`approximate` (exact / sgamma / slognorm), **m** `entropy_fit`, `fzapprox`

11. Meta

m `help`, **m** `json` (persist), `value_type`, `spec`, `spec_ex`

Notes:

[0]: Full signature also has `sev_signed/reflect`, `agg_premium/shift/reflect` (the pnl affine wrapper), `reins_bucket`, `dsev_bucket`; see help.

[1]: matches Portfolio.

Abbreviations: gc_n = gross (subject), ced_{ed}, net; m=mean, cv=coefficient of variation, sd=standard deviation, var=variance, skew(ness); VaR=value-at-risk; est=FFT estimate.